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Exercícios - 3

Controle de Saída Digital com PWM no ESP32 no Simulador Wokwi

Diagrama em blocos

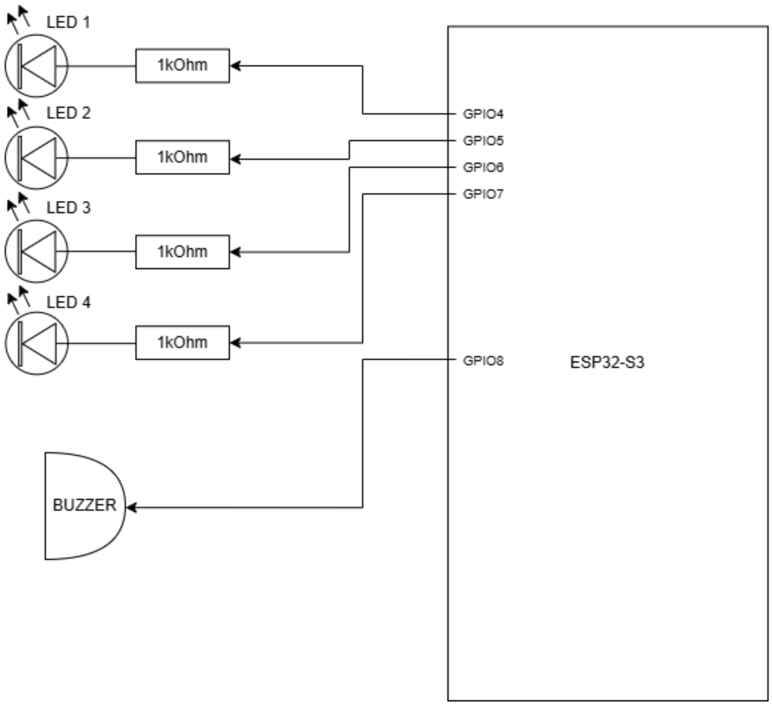
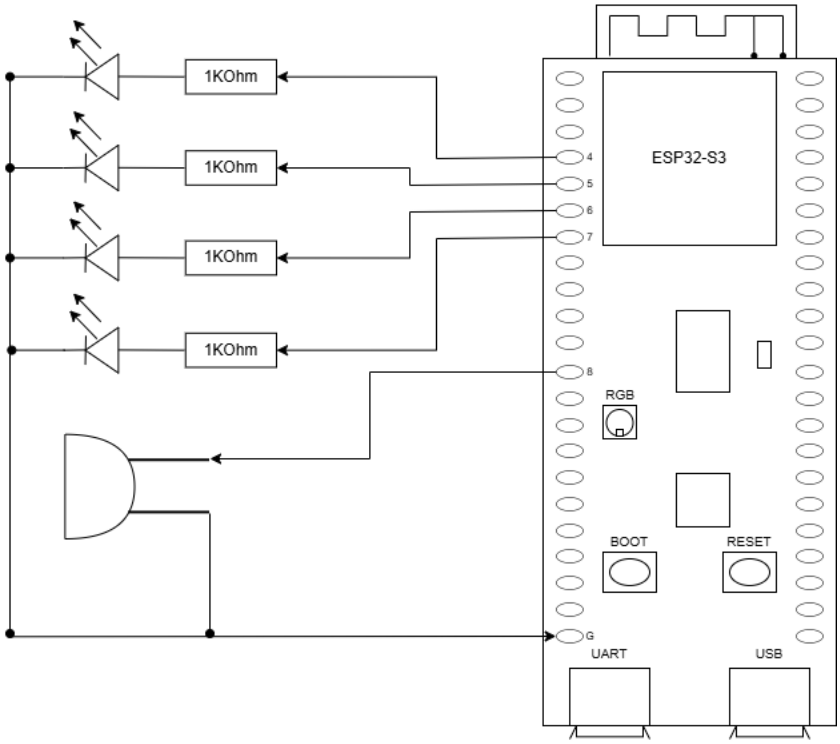
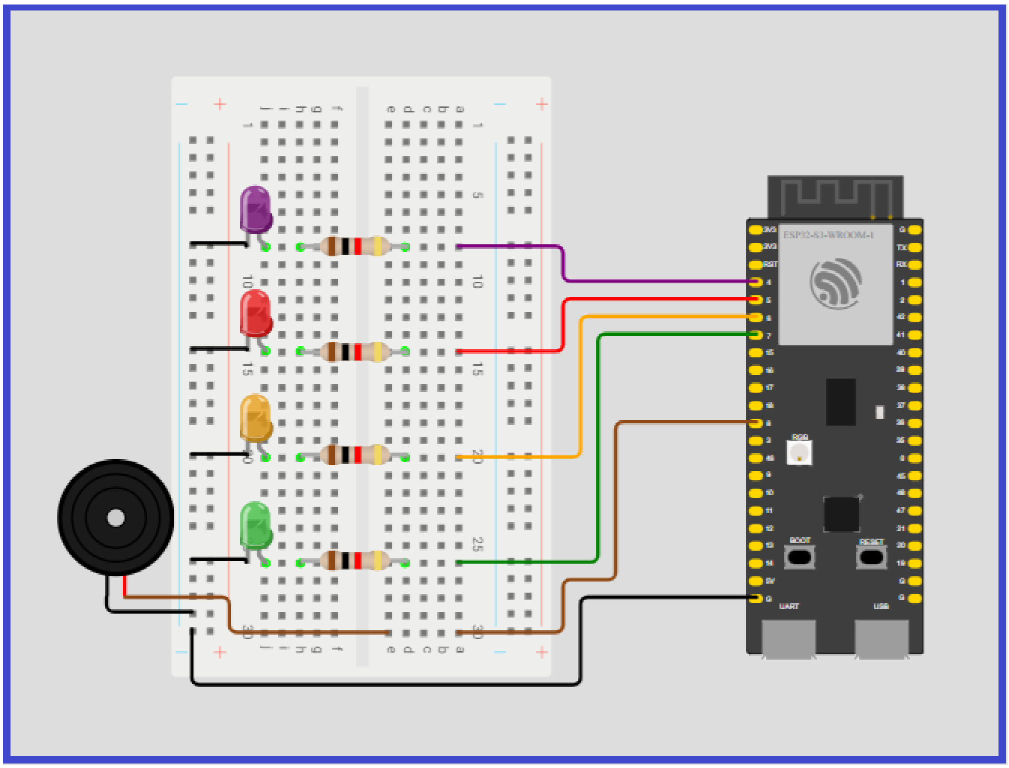


Diagrama esquemático



Simulação do circuito no Wokwi



Meu circuito no Wokwi

Para o desenvolvimento do código em C para circuito no Wokwi.

```
#include <stdio.h>
#include "freertos/FreeRTOS.h"
#include "freertos/task.h"
#include "driver/ledc.h"

#define LED1_GPIO    4
#define LED2_GPIO    5
#define LED3_GPIO    6
#define LED4_GPIO    7
#define BUZZER_GPIO 8

#define LEDC_MODE         LEDC_LOW_SPEED_MODE
#define LED_TIMER         LEDC_TIMER_0
#define BUZZER_TIMER      LEDC_TIMER_1

#define LED_RES           LEDC_TIMER_10_BIT
```

```c
#define DUTY_MAX        ((1 << 10) - 1)  // 1023

#define FREQ_MIN        500
#define FREQ_MAX        2000

void pwm_init() {
    // Configuração Timer LEDs
    ledc_timer_config_t led_timer = {
        .speed_mode = LEDC_MODE,
        .timer_num = LED_TIMER,
        .duty_resolution = LED_RES,
        .freq_hz = 5000,
        .clk_cfg = LEDC_AUTO_CLK
    };
    ledc_timer_config(&led_timer);

    // Configuração Timer Buzzer
    ledc_timer_config_t buzzer_timer = {
        .speed_mode = LEDC_MODE,
        .timer_num = BUZZER_TIMER,
        .duty_resolution = LED_RES,
        .freq_hz = FREQ_MIN,
        .clk_cfg = LEDC_AUTO_CLK
    };
    ledc_timer_config(&buzzer_timer);

    // Configuração Canais LEDs (0 a 3)
    int led_gpios[] = {LED1_GPIO, LED2_GPIO, LED3_GPIO, LED4_GPIO};
    for (int i = 0; i < 4; i++) {
        ledc_channel_config_t ch = {
            .gpio_num = led_gpios[i],
            .speed_mode = LEDC_MODE,
            .channel = i,
            .timer_sel = LED_TIMER,
            .duty = 0,
            .hpoint = 0
        };
        ledc_channel_config(&ch);
    }

    // Configuração Canal Buzzer (Canal 4)
```

```c
    ledc_channel_config_t buzzer_ch = {
        .gpio_num = BUZZER_GPIO,
        .speed_mode = LEDC_MODE,
        .channel = LEDC_CHANNEL_4,
        .timer_sel = BUZZER_TIMER,
        .duty = 0,
        .hpoint = 0
    };
    ledc_channel_config(&buzzer_ch);
}

void app_main(void) {
    pwm_init();

    int32_t duty = 0; // Usando int32 para evitar erros de overflow
    int direction = 1;
    int step = 20;

    while (1) {
        // Atualiza Brilho dos LEDs
        for (int ch = 0; ch < 4; ch++) {
            ledc_set_duty(LEDC_MODE, ch, (uint32_t)duty);
            ledc_update_duty(LEDC_MODE, ch);
        }

        // Atualiza Frequência do Buzzer (Som sobe e desce)
        // Mapeamento linear: frequencia segue o duty dos LEDs
        uint32_t freq = FREQ_MIN + ((FREQ_MAX - FREQ_MIN) * duty / DUTY_MAX);
        ledc_set_freq(LEDC_MODE, BUZZER_TIMER, freq);

        // Volume do buzzer em 50% enquanto houver brilho nos LEDs
        // Se o duty for 0, desligamos o buzzer para silêncio total no ciclo
        uint32_t buzzer_vol = (duty > 0) ? (DUTY_MAX / 2) : 0;
        ledc_set_duty(LEDC_MODE, LEDC_CHANNEL_4, buzzer_vol);
        ledc_update_duty(LEDC_MODE, LEDC_CHANNEL_4);

        // Lógica de Oscilação (Subida e Descida)
        duty += (direction * step);

        if (duty >= DUTY_MAX) {
            duty = DUTY_MAX;
```

```
            direction = -1; // Começa a descer
        }
        else if (duty <= 0) {
            duty = 0;
            direction = 1; // Começa a subir
        }

        vTaskDelay(pdMS_TO_TICKS(20));
    }
}
```

Com o Wokwi, foi gerado um arquivo .json com o código abaixo, representando a montagem da simulação do circuito.


```
{
  "version": 1,
  "author": "Anonymous maker",
  "editor": "wokwi",
  "parts": [
    {
      "type": "wokwi-breadboard-half",
      "id": "bb2",
      "top": -99.9,
      "left": -246.3,
      "rotate": 90,
      "attrs": {}
    },
    {
      "type": "board-esp32-s3-devkitc-1",
      "id": "esp",
      "top": -105.78,
      "left": 129.37,
      "attrs": { "builder": "esp-idf" }
    },
    {
      "type": "wokwi-led",
      "id": "led1",
      "top": 63.6,
      "left": -159.4,
      "attrs": { "color": "limegreen" }
    },
```

```
{
  "type": "wokwi-led",
  "id": "led2",
  "top": 5.1,
  "left": -158.9,
  "attrs": { "color": "orange" }
},
{
  "type": "wokwi-led",
  "id": "led3",
  "top": -52.5,
  "left": -158.9,
  "attrs": { "color": "red" }
},
{
  "type": "wokwi-resistor",
  "id": "r1",
  "top": 99.05,
  "left": -114.7,
  "attrs": { "value": "1000" }
},
{
  "type": "wokwi-resistor",
  "id": "r2",
  "top": 41.45,
  "left": -114.7,
  "attrs": { "value": "1000" }
},
{
  "type": "wokwi-resistor",
  "id": "r3",
  "top": -15.25,
  "left": -115.2,
  "attrs": { "value": "1000" }
},
{
  "type": "wokwi-buzzer",
  "id": "bz1",
  "top": 40.8,
  "left": -247.8,
  "attrs": { "volume": "0.1" }
```

```
    },
    {
      "type": "wokwi-resistor",
      "id": "r4",
      "top": -72.85,
      "left": -115.2,
      "attrs": { "value": "1000" }
    },
    {
      "type": "wokwi-led",
      "id": "led4",
      "top": -109.2,
      "left": -159.4,
      "attrs": { "color": "purple" }
    }
  ],
  "connections": [
    [ "esp:TX", "$serialMonitor:RX", "", [] ],
    [ "esp:RX", "$serialMonitor:TX", "", [] ],
    [ "esp:GND.1", "bb2:bn.25", "black", [ "h-96.05", "v48", "h-212.5" ] ],
    [ "led2:C", "bb2:bn.16", "black", [ "v0" ] ],
    [ "led3:C", "bb2:bn.11", "black", [ "v0" ] ],
    [ "bb2:bn.21", "led1:C", "black", [ "h0" ] ],
    [ "esp:5", "bb2:14t.a", "red", [ "h-105.65", "v9.6" ] ],
    [ "esp:6", "bb2:20t.a", "orange", [ "h-96.05", "v67.2" ] ],
    [ "esp:7", "bb2:26t.a", "green", [ "h-86.45", "v9.6" ] ],
    [ "esp:4", "bb2:8t.a", "purple", [ "h-105.65", "v-19.2" ] ],
    [ "led4:C", "bb2:bn.6", "black", [ "v0" ] ],
    [ "bz1:1", "bb2:bn.24", "black", [ "v0" ] ],
    [ "bz1:2", "bb2:30t.e", "#8f4814", [ "h57.2", "v19.2" ] ],
    [ "bb2:30t.a", "esp:8", "#8f4814", [ "h57.6", "v-28.8", "h28.8", "v-86.4" ]
    [ "led1:A", "bb2:26b.j", "", [ "$bb" ] ],
    [ "led2:A", "bb2:20b.j", "", [ "$bb" ] ],
    [ "led3:A", "bb2:14b.j", "", [ "$bb" ] ],
    [ "r1:1", "bb2:26b.h", "", [ "$bb" ] ],
    [ "r1:2", "bb2:26t.d", "", [ "$bb" ] ],
    [ "r2:1", "bb2:20b.h", "", [ "$bb" ] ],
    [ "r2:2", "bb2:20t.d", "", [ "$bb" ] ],
    [ "r3:1", "bb2:14b.h", "", [ "$bb" ] ],
    [ "r3:2", "bb2:14t.d", "", [ "$bb" ] ],
    [ "led4:A", "bb2:8b.j", "", [ "$bb" ] ],
```

```
    [ "r4:1", "bb2:8b.h", "", [ "$bb" ] ],
    [ "r4:2", "bb2:8t.d", "", [ "$bb" ] ]
  ],
  "dependencies": {}
}
```